(P3) $\vec{E} = k\left(1-e^{-\alpha r}\right)\frac{\hat{r}}{r^2}$

$$\oint \vec{E}\cdot\hat{n}\,da = \frac{q}{\varepsilon_0} \Rightarrow q(r) = \varepsilon_0\, k\left(1-e^{-\alpha r}\right)\frac{\hat{r}}{r^2}\cdot 4\pi r^2$$

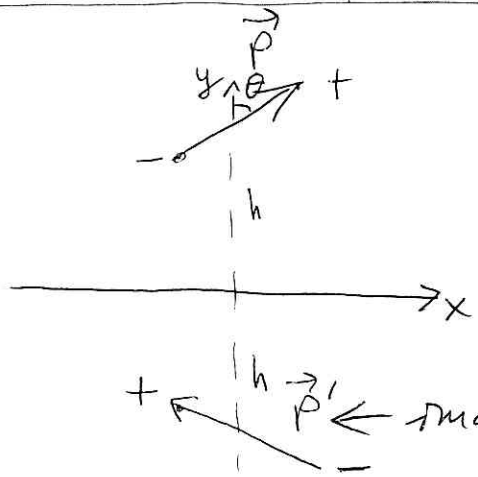
$$= 4\pi\varepsilon_0\, k\left(1-e^{-\alpha r}\right)$$

$\Rightarrow$ charge with sphere of radius $r=\frac{1}{\alpha}$ is

$$q\left(\tfrac{1}{\alpha}\right) = 4\pi\varepsilon_0\, k\left(1-\frac{1}{e}\right).$$

(P4)

a)



$p'_x = -p_x$, $p_x = p\sin\theta$

$p'_y = p_y$, $p_y = p\cos\theta$

$$\vec{E}_{dip}(\vec{r}) = \frac{3\vec{r}(\vec{r}\cdot\vec{p}) - \vec{p}\,r^2}{r^5}$$

$\frac{1}{2}$ is specific to image problem: shifting $\vec{p}$ by δh increases the distance to the image dipole by $2\delta h$!

$U_{dip} = -\frac{1}{2}\vec{p}\cdot\vec{E}'$ where $\vec{E}'$ is el. field due to image dipole $\vec{p}\,'$

$$U = -\frac{1}{2}\left(p_x\hat{x} + p_y\hat{y}\right)\cdot\frac{3\,2h\hat{y}\,(2h\hat{y}\cdot\vec{p}\,') - \left(p'_x\hat{x} + p'_y\hat{y}\right)(2h)}{(2h)^5}$$

$$= -\left(p_x\hat{x} + p_y\hat{y}\right)\cdot\frac{3\hat{y}\,p'_y - \left(p'_x\hat{x} + p'_y\hat{y}\right)}{2\cdot(2h)^3}$$

$$= -\left(p_x\hat{x} + p_y\hat{y}\right)\cdot\frac{\left(2p'_y\hat{y} - p'_x\hat{x}\right)}{2\cdot(2h)^3}$$

$$= \frac{1}{2\cdot(2h)^3}\left(p_x p'_x - 2p_y p'_y\right) = \frac{-p_x^2 - 2p_y^2}{16h^3}$$

$$= \frac{-p^2}{16h^3}\left(\sin^2\theta + 2\cos^2\theta\right) = -\frac{(qd)^2}{16h^3}\left(1 + \cos^2\theta\right)$$

Energy is minimized by $\cos\theta_0 = \pm 1 \Rightarrow \theta_0 = 0, \pi$.

b) Kinetic energy of dipole $K = \frac{1}{2} I \dot{\theta}^2$

where I is the moment of inertia.

$$L = \frac{1}{2} I \dot{\theta}^2 + \frac{p^2}{16h^3}(1+\cos^2\theta)$$

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{\theta}} - \frac{\partial L}{\partial \theta} = 0$$

$$I\ddot{\theta} = \frac{p^2}{16h^3}\, 2\cos\theta(-\sin\theta) = -\frac{p^2}{16h^3}\sin 2\theta$$

$\theta = \theta_0 + \delta\theta$, $\sin 2\theta = \sin(2\theta_0 + 2\delta\theta) \approx 2\delta\theta$ for both $\theta_0 = 0$ and $\theta_0 = \pi$.

$$\Rightarrow \delta\ddot{\theta} = -\frac{p^2}{8h^3 I}\delta\theta$$

$$\Rightarrow \Omega^2 = \frac{p^2}{8h^3 I}$$